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MILITARY STANDARD

SURVEILLANCE INSPECTION OF INDUSTRIAL STOCKS OF BULK PROPELLANT IN STORAGE

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ARMED FORCES SUPPLY SUPPORT CENTER

WASHINGTON 25, D. C.

Surveillance Inspection of Industrial Stocks of Bulk Propellant in Storage

MIL-STD-1231B 2 August 1962

1. This standard has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force, effective 2 August 1962.

2. Recommended corrections, additions, or deletions should be addressed to the Standardization Division, Armed Forces Supply Support Center, Washington 25, D. C.

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1. SCOPE

1.1 SCOPE. This standard establishes inspection procedures and techniques for annual inspection and special inspection of bulk propellant by lot in storage not scheduled for shipment, acceptance testing, or loading.

1.2 APPLICATION. This standard applies to each installation where industrial stocks of bulk propellant are stored.

2. REFERENCED DOCUMENTS

(Not Applicable)

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3. DEFINITIONS

The following definitions apply to specific terms used in this standard.

3.1 SURVEILLANCE. Inspection procedures to be followed for the purpose of determining the initial satisfactory condition, the year to year change and the level of stability of stored bulk propellant.

3.2 BULK PROPELLANTS. Propellants lying between finished manufacture and assembly into rounds; including loose propellant, propelling charges packed in round metal drums, mortar increments and rocket grains (noncomposite) ; but not including in-process propellant (for example; base grain for cast propellant that is programmed unless 5 years or older.

3.3 TECHNIQUES. The method of inspection described in this standard is termed the propellant quick test. This test replaces the N/10 Methyl-Violet Paper Test.

3.4 HIGHER AUTHORITY. Higher authority shall be Industrial ((ORDLY-AI) or Field Service (ORDLY-Q) office at the Ordnance Ammunition Command, whichever is applicable.

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4. GENERAL REQUIREMENTS OR STATEMENTS

4.1 POLICIES. At Government-owned contractor operated installations, surveillance work may be performed by contractor personnel under the technical supervision of Government inspection personnel. Government inspection personnel will perform surveillance at all other facilities.

4.2 RESPONSIBILITIES. Each installation at which industrial stocks of bulk propellant are stored will:—

4.2.1 Utilize the techniques described in this standard to perform surveillance examination of bulk propellant.

4.2.2 Report results of surveillance examination when those results indicate that the stability level of the propellant has undergone a significant decrease.

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5. DETAILED REQUIREMENTS OR STATEMENTS

5.1 TEST PROCEDURES (PREFACE). The propellant quick test is based on a standard-laboratory test for nitrates. A mixture of diphenylamine and concentrated sulfuric acid will become blue in the presence of nitrates. On dilution, yellow-green-blue components appear. This test has been adapted to the detection of oxides of nitrogen, in the field, by pre-impregnating strips of filter paper with diphenylamine from a 1 percent solution in alcohol. These strips are spot-wetter at the test site with 5 percent sulfuric acid. On 1- or 2-hour exposure to the propellant atmosphere, the wetted area will react with the decomposition products, taking on a color which is within 8 distinguishable color stages.

5.1.1 *Materials.*

Quick Test Color Chart

Quick Test Papers, in packet (procured for the current test season).

Acid reactant (5 per cent sulfuric acid (H_2SO_4) by volume) in dispenser bottle, renewed annually.

Drop-dispenser or glass rod.

Bottle of sodium bicarbonate solution for neutralizing acid spills.

Timer.

Metal twist-on tags, or equal.

Worksheets.

Standard inspection gear, including electric light headgear, nonsparking tool for lifting propellant container lid, and protective eye equipment.

5.1.2 *Test temperature.*

5.1.2.1 The test shall be conducted when the prevailing temperature within the storage site or temporary test site is 70° ± 50° F. (For situations requiring exceptions, see paragraph 5.1.11.1 and following.)

5.1.2.2 The test temperature shall be considered as that determined midway of the propellant stack.

5.1.3 *Lot representation.*

5.1.3.1 *Initial test.*

5.1.3.1.1 *Basic representation.* 1 percent but not less than 10 containers.

5.1.3.1.2 *Control representation.* 10 containers of the above distinctly tagged as "Controls."

5.1.3.1.3 *Additional representation.*

On production of a Stage 2, 1 percent additional.

On production of a Stage 4, 2 percent additional.

On production of a Stage 6, 4 percent additional with evaluation of one-fourth of these at the recheck time intervals prescribed in paragraph 5.1.8.2.

5.1.3.1.4 *Noncontrol representation.* All containers tested other than the controls, to be distinctively tagged for nonrepeat testing until all containers in the lot have been tested.

5.1.3.2 *Subsequent annual tests.*

5.1.3.2.1 *Representation.* Control containers (10) plus 1 percent of the lot.

5.1.3.2.2 *Additional representation.* As listed in paragraph 5.1.3.1.3.

5.1.3.2.3 *Noncontrol representation.* As listed in paragraph 5.1.3.1.4.

5.1.4 *Test routine.* The test routine shall consist of the immediately progressive accomplishment at the test site of the following (par. 5.1.5 through 5.1.9):

5.1.5 *Preparation: Propellant containers.*

5.1.5.1 Make the random percentage selection of the containers so as to effect gen-

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eral representation of the lot—center front and rear, and extremes of the space occupied by the lot; low, medium, and high in the stack.

5.1.5.2 Tag the containers so selected.

5.1.5.3 Loosen the spiders or plugs, *but not the lids.*

5.1.5.4 Remove the lid from one container, other than the sample, to ascertain if the rim of the lid seating may force the test paper back up against the lid, on closing (see par. 5.1.7.3).

5.1.5.5 For cases involving threaded plug openings, see options in appendix A.

5.1.6 *Preparation: Quick test papers.*

5.1.6.1 Select from the packet enough strips for immediate need. (Strips are in groups of 15, lightly attached at the top. The lower half of the strip is impregnated with the indicator, diphenylamine. Strips must be handled by the tops.) Replace unneeded strips and reclose packet.

5.1.6.2 Deposit a touch of the acid reactant about 1 inch from the bottom of each strip. The acid should make a spot about 3/16 inch in diameter. Too much wetting will cause the diphenylamine to "run" to the outer edge, and the end point will be uneven. For ease in handling, the individual strips should not be detached until ready for use.

5.1.6.3 Air-dry the strips by gentle waving for 3 to 5 minutes, in order to assure dispersal of the reactant among the paper fibers.

5.1.7 *Test paper insertion.*

5.1.7.1 "Crack" the lid of the propellant container by carefully prying up both lid and gasket on the convenient side. A "crack" no higher than 1/4-inch should permit easy insertion of the paper strip. If the gasket springs so as to make reclosing difficult, reject the container for testing, reclose, and substitute another container.

5.1.7.2 Separate the strips by tearing up through the top, in line with the pre-cut, to assure full length of the strip.

5.1.7.3 If paragraph 5.1.5.4 has shown that the test paper may be forced back against the lid on closing, impart a downward curl to the upper middle of the strip by drawing it at a sharp angle over a pencil for half of its length, prior to insertion.

5.1.7.4 Slip the test paper under the lid so that no greater than 1 inch will protrude and reseat the lid with hand pressure. For insertion of test strips in threaded-plug openings, see appendix A.

5.1.8 *Test time.*

5.1.8.1 Standard test time is 1 hour ± 5 minutes except in the testing of cloth bagged charges or bagged increments, when it is 2 hours ± 10 minutes. This is the elapsed time from the insertion of the paper to the removal of the paper from each container.

5.1.8.2 Recheck test times (to be used when a stage 6 is produced during the standard test time) are portions of a standard test time. To accomplish this, insert 4 strips in each container of the prescribed number and withdraw one each at 5, 10, 15, and 30 minutes when the standard test time is 1 hour, and at 10, 20, 30, and 60 minutes when the standard test time is 2 hours.

5.1.9 *Evaluation of paper test color.*

5.1.9.1 Withdraw a test strip.

5.1.9.2 Immediately compare the color of the wetted area with the color of the stages on the color chart.

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5.1.9.3 For ease of comparison, fold the test paper across the middle of the wetted spot and bring the discolored fold against the swatches on the color chart.

5.1.9.4 Identify as the test value that stage of the charted color which is reasonably approached by the test color spot.

5.1.9.5 Identify the test value, when the color of the test spot lies definitely between charted stages, by citing the boundary stages separated by a slash. (For example: Stage 0/1, 2/3 etc.)

5.1.10 *Recording of data.*

5.1.10.1 Maintain permanent record of test data on Depot Surveillance Record, 00 Form 1071.

5.1.10.2 List the quantity of containers of each stage yielded. In subsequent annual tests, report the data on the control containers separately.

5.1.10.3 Record the test temperature (see par. 5.2.1.2).

5.1.11 *Special measures.*

5.1.11.1 *Test temperature options.*

5.1.11.2 Options in temperature at time of test are permitted only when (1) the prescribed range of 70° ± 5° F. is not adequately characteristic of the region, or (2) manpower is not adequate for complete annual testing within the time-span of normal temperature in that range.

5.1.11.3 A test-temperature range of 60° ± 5° F. is permitted; but for relief only under the above circumstances.

5.1.11.4 Use of this relief measure entails a modification of test procedure, as given in paragraph 5.1.3.1.3, "Additional Representation" as follows:

On production of a stage 2, 1 percent additional.

On production of a Stage 3, 2 percent additional.

On production of a stage 4, 4 percent additional with evaluation of 1 percent of these at the recheck time intervals prescribed in paragraph 5.1.8.2.

5.1.11.5 The CONUS climate is such that test-temperature relief above the prescribed 70° ± 5° F. should not be required except in cases of an emergency nature.

5.1.11.6 A test-temperature range of 80° ± 5° F. may be used for such emergency situations.

5.1.11.7 No changes in the basic instructions are required under the higher test-temperature; but all resultant data shall be submitted to the next higher authority for evaluation.

5.1.11.8 *Nontestable packaging.*

5.1.11.9 Mortar sheet increments in nontestable packaging shall be tested by use of a lot-control sample in accordance with the following:

5.1.11.9.1 Weight—2 pounds minimum.

5.1.11.9.2 Selection—from a minimum of 5 representative boxes.

5.1.11.9.3 Condition—decellophaned.

5.1.11.9.4 Containers—Air-tight container of such size that it will be filled within 3 inches of the top.

5.1.11.9.5 Positioning—Midway within the area occupied by the lot.

5.1.11.9.6 *Deviations.* Improvements in control through reasonable increase in sample weight (5.1.11.9.1) or lot representation (5.1.11.9.2), but without alteration of the sample-to-container ratio (5.1.11.9.4) shall

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be proposed to the next higher authority for approval.

5.1.11.10 Rocket grains ((noncomposite) in nontestable packaging shall be tested by use of a lot-control sample in accordance with the following:

5.1.11.10.1 Weight—Not applicable.

5.1.11.10.2 Representation—5 grains from 5 representative boxes.

5.1.11.10.3 Condition—Unwrapped; cross-cut into pieces with a length of twice the diameter of the grain.

5.1.11.10.4 Container—Airtight container of such size that it will accommodate the closely packed sample to within 3 inches of the top.

5.1.11.10.5 Positioning — Midway within the area occupied by the lot.

5.1.11.10.6 Deviations — Improvements in control through reasonable increase in lot representation (5.1.11.10.2) or size of rocket grain portions (5.1.11.10.3), but without alteration of the sample-to-container ratio (5.1.11.10.4) shall be proposed to the next higher authority for approval.

5.1.11.11 Miscellaneous propellants (ignition powders, ECBF, etc.), in nontestable packaging shall be tested by use of applicable lot-sample technique (par. 5.1.11.9 or 5.1.11.10); or specific instructions shall be requested from the next higher authority.

5.1.12 *Safety factors.*

5.1.12.1 All actions involved in this routine, or related thereto, shall be in accordance with established safety procedures.

5.1.12.2 When the acid reactant is prepared by the installation, the acid must be added slowly to the water with constant stirring.

5.1.12.3 The acid reactant is mildly irritating to the skin. Handle the acid bottle and dispenser with care; avoid contact. If contact occurs, wash liberally with neutralizing solution or water.

5.1.12.4 Contact of the acid reactant with the eyes might be serious. The wearing of light protective eye equipment for the wetting operations is warranted. If contact occurs, flush liberally with water; report immediately to the dispensary.

5.1.13 *Clean-up.*

5.1.13.1 Replace and tighten all spiders or plugs.

5.1.13.2 Destroy all used test papers.

5.1.13.3 Remove the acid reactant from the storage area.

5.1.13.4 Remove the unused test papers from the storage area; keep in a dark place free from chemical vapors.

5.2 **REPORTS.**

5.2.1 Test results consisting of the following shall be reported on the Annual Ammunition Inspection Lot Number Report for each lot tested:

5.2.1.1 Quantity of containers in each stage.

5.2.1.2 Temperature within test site at time of test.

5.2.2 In testing at standard temperatures (70° ± 5° F.) when the results on a lot indicate a stage 6 or beyond, or when a 4-stage jump from the previous year's inspection is noted, the results shall be reported by the most expeditious means to the next higher authority and disposition instructions requested.

5.2.3 In testing at substandard tempera-

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tures (60° ± 5° F.) when the results on a lot indicate a stage 4 or beyond when a 2-stage jump from the previous year's inspection is noted, the results shall be reported by the most expeditious means to the next higher authority and disposition instructions requested.

5.3 DEMILITARIZED PROPELLANT.

5.3.1 When ammunition and/or component items are demilitarized the next higher authority is to be advised immediately as to quantity, visual condition and degree of contamination, i.e., tracer compound, etc., of the bulk propellant. Appropriate disposition and/or surveillance instructions will be provided.

5.4 HAZARDOUS PROPELLANT.

5.4.1 Propellant which has deteriorated to a hazardous degree will yield a stage 8 (blue) reaction in the 15-minute recheck test following the 1-hour standard test or in the 30-minute recheck test following the 2 hour standard test accompanied by a yellowing of the dry portion of the diphenylamine-impregnated test strip. When this condition is encountered, all containers in the lot shall be tested and if 25 percent or more of the lot produces this result, all of that lot of propellant shall be stack wetted, the lot immediately removed, the propellant locally disposed, and higher authority shall be notified of the action taken. If less than 25 percent is found, the containers, which have reacted to a hazardous degree shall be wetted, removed from storage, and the propellant destroyed. Higher authority shall be notified of the condition of the balance of the lot and disposition instructions requested.

5.5 TEST MATERIAL SUPPLY.

5.5.1 Diphenylamine paper and color charts shall be obtained, by letter request from Picatinny Arsenal, Dover, N. J., ATTN: Explosives and Propellant Laboratory. A supply representing only the actual yearly need should be requested. The acid reactant must be procured locally.

5.5.2 Notice: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

5.5.3 Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contractoring officer.

Copies of this standard for military use may be obtained as indicated in the foreword to, or the general provisions of, the Department of Defense Index of Specifications and Standards.

The title and identifying symbol should be stipulated when requesting copies of military standards."

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APPENDIX A

The material in this appendix forms a mandatory part of the standard.

10. Options for Insertion of Propellant Quick Test Strips into Threaded Plug Openings, Metal Drum Propellant Containers.

10.1 Application—The options are applicable to particular situations existing in storage of containers with this type of opening.

10.2 Option: Use of threaded plug.

10.2.1 Applicable to the situation in which the propellant is suitably free from the area of the opening.

10.2.2 Insert the test strip, and reclose with the original threaded plug by a part turn sufficient to hold; or

10.2.3 Attach a shortened test strip to the bottom of the plug with scotch tape, and reclose by hand pressure.

10.3 Option: Use a slotted rubber stopper.

10.3.1 Applicable as in paragraph 10.2.1.

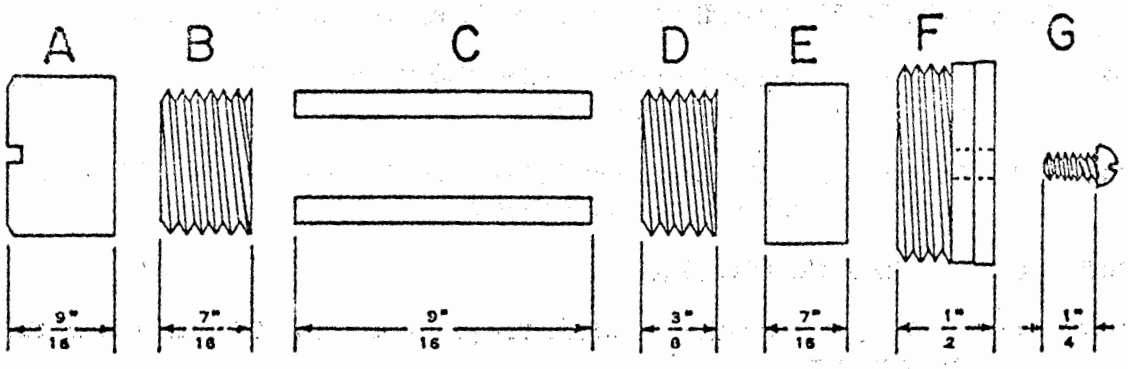
10.3.2 Insert the test strip into the slot in the rubber stopper; compress and insert into the opening.

10.4 Option: Use of adapter (fig. 1).

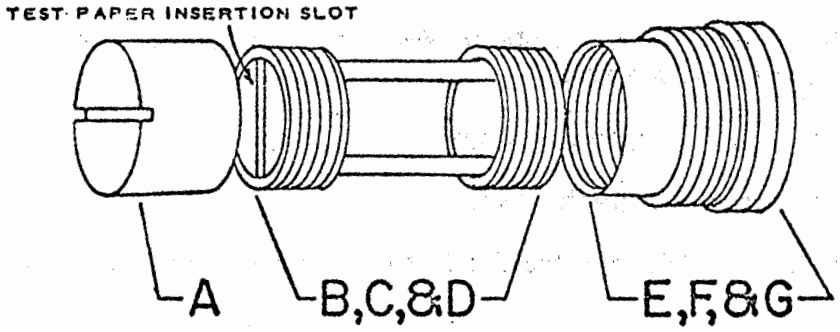
10.4.1 Applicable to the situation in which the paper liner is pressed against the opening.

10.4.2 Insert a shortened test strip in the adapter, spot-wet, and insert the adapter into the container opening.

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VIEW I



VIEW II

VIEW I — COMPONENT PARTS SKETCH (SIDE VIEW)

A & F ..... FEMALE SCREW CAP.
B & D ..... MALE SCREW END.
C ..... BRASS ROD, 1/8" DIAMETER.
F ..... STANDARD DRUM INSPECTION PLUG.
G ..... BRASS MACHINE SCREW, 3/16".

VIEW II — ASSEMBLY SKETCH (OBLIQUE VIEW)

A ..... CAP WHICH HOLDS TEST PAPER IN PLACE.
B, C, & D ... TEST PAPER HOLDER • VIEW SHOWS COMPONENTS B, C, & D ASSEMBLED AND ALSO TEST PAPER INSERTION SLOT.
E, F, & G ... ASSEMBLED VIEW OF CAP E FASTENED TO DRUM INSPECTION PLUG.

REMARKS: THE TEST PAPER HOLDER IS FINALLY INSERTED IN CAP E AND SPOT WELDED TO HOLD IT IN PLACE PERMANENTLY.

FIGURE 1. *Propellant quick test inspection device for use in metal propellant container.*

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